# Department of Artificial Intelligence and Data Science (ADS)

## Green University of Bangladesh

**Semester:** Spring 2026, B.Sc. in ADS

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## " Introduction to Python Programming "

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**Lab Report No:** 02

**Course Title:** Introduction to Computational Thinking and Programming

**Course Code:** ADS 100

**Section:** 261 D2

## Students Details

| Name | ID |
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| Lab Report Status | |
|---|---|
| **Marks:** | **Signature:** |
| **Comments:** | **Date:** |

**Lab Experiment Date:** 28/03/ 26

**Lab Submission Date:** 06 /04/ 25

**Course Teacher's Name:** Wahia Tasnim

# 1. TITLE OF THE LAB REPORT EXPERIMENT

- **Write a Python program to swap two variables without using a third variable.**
- **Write a Python program that takes two string variables and prints their concatenation in two different**

**ways: using + and using a comma in print().**

- **Write a Python program that takes a variable and prints its type using type().**
- **Write a Python program that assigns three different values to three variables in a single line and prints**

**them.**

- **Write a Python program that takes three numbers as input, calculates their sum and average, and prints**

**the result.**

- **Write a Python program to input a number and perform left and right shifts on the number.**

## 2. Objective/Aim :

**The objective of this lab is to develop fundamental problem-solving and programming skills using Python by writing simple programs by using variables, data types, arithmetic operations, user input/output and bitwise operators. Through this exercise, we learn how to manipulate variables, combine strings, determine variable types, perform calculations and apply bitwise operations.**

**Specifically, the aims are:**

1. **To swap two variables without using a third variable.**
2. **To demonstrate string concatenation using ‘ + ’ and ‘ . ’ method.**
3. **To identify the data type of a variable using the type() function.**
4. **To assign multiple variables in a single line.**
5. **To compute the sum and average of three numbers.**
6. **To understand bitwise shift operations in Python.**

**3. PROCEDURE :**

**The algorithm of the 6 problems are given below.**

**Problem 01: Swap two variables without using a third variable**

**Step 1: Input two values into variables a and b.**

**Step 2: Print the current value of a and b.**

**Step 3: Swap a and b by assigning the current value of b to a and the current value of a to b simultaneously.**

**Step 4: Print the swapped values of a and b.**

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**Problem 02: Concatenate two strings**

**Step 1: Take two string inputs from the user.**
**Step 2: Concatenate the strings using the + operator.**
**Step 3: Print the result.**
**Step 4: Print the two strings again using a comma inside the print() function.**

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**Problem 03: Print variable type**

**Step 1: Take an input value from the user and store it in a variable.**
**Step 2: Use the type() function to identify the data type of the variable.**
**Step 3: Print the result showing the data type.**

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**Problem 04: Assign three variables in a single line**

**Step 1: Assign three different values to variables a, b, and c in a single line using comma separation.**
**Step 2: Print the values of the variables.**

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**Problem 05: Calculate sum and average**

**Step 1: Take three numbers as input from the user.**
**Step 2: Store the values in variables a, b, and c.**
**Step 3: Calculate the sum using sum = a + b + c.**

**Step 4: Calculate the average using average = sum / 3.**
**Step 5: Print both results.**

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**Problem 06: Perform left and right shift**
**Step 1: Take an integer input from the user.**
**Step 2: Apply the left shift operator (<<) to shift the bits of the number to the left..**
**Step 3: Apply the right shift operator (>>) to shift the bits of the number to the right.**
**Step 4: Print the results of both operations.**

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## 4. TEST RESULT / OUTPUT:

Problem01:

**Swap two variables without using a third variable:**

**Input:**

```python
a = input("Enter first value a: ")
b = input("Enter second value b: ")

print(f"Current values, a: {a}, b: {b}")

a, b = b, a

print(f"After swapping: ")
print(f"New values, a: {a}, b: {b}")
```

**Output:**

```
uv run lab02_p1_swap_variables.py
Enter first input a: 1
Enter second input b: 2
Current Inputs, a: 1, b: 2
After swaping:
New Inputs, a: 2, b: 1
```

Problem02:

**Takes two string variables and prints their concatenation in two different ways: using + and using a comma in print():**

**Input:**

```python
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

print("Using + :", str1 + str2)
print("Using , :", str1, str2)
```

**Output:**

```
uv run lab02_p2_string_concatenation.py
Enter first string: hello
Enter second string: world
Using + : helloworld
Using , : hello world
```

Problem03:

**Takes a variable and prints its type using type().:**

**Input:**

```python
x = input("Enter a value: ")

print("The type of the variable is: ", type(x))
```

**Output:**

```
uv run lab02_p3_print_data_type.py
Enter a value: 3
The type of the variable is:  <class 'str'>
```

Problem04:

**Assign three different values to three variables in a single line and prints them.:**

**Input:**

```python
a, b, c = 10, 20, 30
print(a, b, c)
```

**Output:**

```
uv run lab02_p4_assign_three_values.py
10 20 30
```

Problem05:

**Take three numbers as input, calculates their sum and average, and prints the result:**

**Input:**

```python
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

sum_numbers = a + b + c
average = sum_numbers / 3

print("The sum is: ", sum_numbers)
print("The average is: ", average)
```

**Output:**

```
uv run lab02_p5_sum_average.py
Enter first number: 20
Enter second number: 30
Enter third number: 60
The sum is:  110.0
The average is:  36.666666666666664
```

Problem06:

**Input a number and perform left and right shifts on the number:**

**Input:**

```python
num = int(input("Enter a number: "))

left_shift = num << 1
right_shift = num >> 1

print("Left shift: ", left_shift)
print("Right shift: ", right_shift)
```

**Output:**

```
uv run lab02_p6_bitwise_shift.py
Enter a number: 8
Left shift:  16
Right shift:  4
```

# 5. Key Python Concepts Used

During the implementation of these programs, several important Python programming concepts were applied:

**Variables**
Variables were used to store input values and intermediate results during computations.

**Multiple Variable Assignment**
Python allows assigning multiple variables in a single line which is helpful for swapping variables and improves efficiency.

**Arithmetic operators**
Operators such as +, -, /, and * were used to perform mathematical calculations such as sum and average.

**String Operations**
String concatenation demonstrated using both the + operator and comma separated output in the print() function.

**Built-in Functions**
Functions like input(), print() and type() were used to take user input, display results and determine the data type of variables.

**Bitwise Operators**
The left shift (<<) and right shift (>>) operators were used to demonstrate binary-level manipulation of numbers.

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# 6. Conclusion

The Lab helped me understand the basic concepts of Python programming through several small programming tasks. By solving these problems, I practiced working with variables, data types, arithmetic operations, strings and bitwise operators.

In the first problem, I learned how two variables can be swapped without using a third variable by applying simultaneous assignment using tuple unpacking that Python allows. This process exchanges the values of the variables in one statement. By using assignment operator like this makes the code efficient and readable and allows swapping of two variables of any data type.

The second problem demonstrated two different ways of joining strings in Python. Using the + operator combines two strings into one, while using a comma in the print() function prints them with a space between them. This showed that python often provides multiple ways to perform the same task.

The third program I used the type() function to display the data type of a variable. This helped me understand how Python treats input values and why type conversion is sometimes necessary when working with numbers.

The fourth problem showed how Python allows multiple variables to be assigned values ina single line. This makes the code shorter and easier to read.

In the fifth problem, I calculated the sum and average of three numbers. This exercise helped reinforce the use of arithmetic operations and user input in Python programs.

Finally, the sixth problem introduced bitwise shift operators. The left shift operator moves bits to the left and effectively multiplies the number by two, while the right shift operator moves bits to the right and divides the number by two. This helped me understand how numbers are processed at the binary level.

Overall, this lab improved my understanding of basic of Python syntax and helped me practice writing simple programs. These exercises strengthened my logical thinking and provided a good foundation for leaning more advanced programming concepts in the future.

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